

data_formatting_pt2_bb

August 6, 2024

1 Creation of the Finalized Cleaned Dataset with Targets

```
[ ]: # Data analysis
import pandas as pd
import numpy as np

# Plotting and Correlation Maths
import seaborn as sns
import scipy as sci

# Simple model development
import sklearn as sk

import matplotlib.pyplot as plt

import plotly as px

import os
```

```
[ ]: df = pd.read_csv('../data/interim/merged_df.csv')
df
```

```
[ ]:
```

	date	name	tempmax	tempmin	temp	feelslikemax	\
0	2018-01-07	Allegheny County	26.2	-4.2	10.1	18.1	
1	2018-01-09	Allegheny County	35.8	28.0	32.5	34.7	
2	2018-01-14	Allegheny County	15.3	2.5	9.2	14.6	
3	2018-01-14	Allegheny County	15.3	2.5	9.2	14.6	
4	2018-01-14	Allegheny County	15.3	2.5	9.2	14.6	
...	...	...	...	...	...	...	
5308	2024-04-13	Allegheny County	59.2	44.5	51.1	59.2	
5309	2024-04-13	Allegheny County	59.2	44.5	51.1	59.2	
5310	2024-04-13	Allegheny County	59.2	44.5	51.1	59.2	
5311	2024-04-14	Allegheny County	81.1	45.0	64.1	79.9	
5312	2024-04-14	Allegheny County	81.1	45.0	64.1	79.9	

```
feelslikemin feelslike dew humidity ... age_range race \
```

0	-16.6	0.6	-3.8	57.2	...	30 - 39	BLACK
1	24.1	26.8	26.6	78.9	...	40 - 49	WHITE
2	-6.0	3.5	0.5	68.0	...	40 - 49	WHITE
3	-6.0	3.5	0.5	68.0	...	20 - 24	WHITE
4	-6.0	3.5	0.5	68.0	...	30 - 39	WHITE
...	...	...	...	...	...	...	...
5308	37.4	47.9	33.0	51.5	...	20 - 24	WHITE
5309	37.4	47.9	33.0	51.5	...	20 - 24	WHITE
5310	37.4	47.9	33.0	51.5	...	20 - 24	WHITE
5311	39.9	62.9	39.5	42.6	...	20 - 24	WHITE
5312	39.9	62.9	39.5	42.6	...	30 - 39	BLACK

	ethnicity_desc	accidental_exposure	victim_od_drug_id	\
0	Unknown	N	3288	
1	NON-HISPANIC OR NOT LATINO	N	527	
2	NON-HISPANIC OR NOT LATINO	N	800	
3	NON-HISPANIC OR NOT LATINO	N	804	
4	NON-HISPANIC OR NOT LATINO	N	799	
...	...	...	...	
5308	NON-HISPANIC OR NOT LATINO	N	55826	
5309	NON-HISPANIC OR NOT LATINO	N	55825	
5310	NON-HISPANIC OR NOT LATINO	N	55825	
5311	NON-HISPANIC OR NOT LATINO	N	55849	
5312	NON-HISPANIC OR NOT LATINO	N	55827	

	susp_od_drug_desc	naloxone_administered	administration_id	survive	\
0	HEROIN	N	0	Y	
1	HEROIN	Y	244	Y	
2	HEROIN	Y	454	Y	
3	HEROIN	Y	457	Y	
4	HEROIN	Y	452	Y	
...	...	...	...	...	
5308	HEROIN	Y	25971	Y	
5309	FENTANYL	Y	25971	Y	
5310	FENTANYL	Y	25972	Y	
5311	HEROIN	Y	25981	Y	
5312	HEROIN	Y	25973	U	

	dose_unit
0	0
1	0
2	0
3	0
4	0
...	...
5308	4
5309	4

```
5310      4
5311      4
5312      4
```

```
[5313 rows x 38 columns]
```

```
[ ]: df.columns
```

```
[ ]: Index(['date', 'name', 'tempmax', 'tempmin', 'temp', 'feelslikemax',
          'feelslikemin', 'feelslike', 'dew', 'humidity', 'precip', 'precipcover',
          'preciptype', 'snow', 'snowdepth', 'cloudcover', 'uvindex',
          'severerisk', 'moonphase', 'icon', 'dose_count', 'incident_id',
          'incident_time', 'day', 'incident_county_name', 'incident_state',
          'victim_id', 'gender_desc', 'age_range', 'race', 'ethnicity_desc',
          'accidental_exposure', 'victim_od_drug_id', 'susp_od_drug_desc',
          'naloxone_administered', 'administration_id', 'survive', 'dose_unit'],
          dtype='object')
```

2 Dropping Duplicates

We can see that on the `incident_id` there are some duplicates that exist. These need to be subsetted such that if the victim and the incident match, then it should be dropped keeping the first.

This will allow for instances where multiple drugs are found to be sifted out while preserving multiple people at a single event.

```
[ ]: df.duplicated(subset=['victim_id', 'incident_id'])
```

```
[ ]: 0      False
      1      False
      2      False
      3      False
      4      False
      ...
      5308    True
      5309    True
      5310    True
      5311    False
      5312    False
      Length: 5313, dtype: bool
```

```
[ ]: print(f'There are a total of {sum(df.duplicated(subset=['victim_id',
↪ 'incident_id']))} duplicated values.')
```

There are a total of 1629 duplicated values.

Drop the duplicated values but keep the first of them, since the event did happen.

```
[ ]: df2 = df.drop_duplicates(subset=['victim_id', 'incident_id'], keep='first')
df2
```

```
[ ]:
```

	date	name	tempmax	tempmin	temp	feelslikemax	\
0	2018-01-07	Allegheny County	26.2	-4.2	10.1	18.1	
1	2018-01-09	Allegheny County	35.8	28.0	32.5	34.7	
2	2018-01-14	Allegheny County	15.3	2.5	9.2	14.6	
3	2018-01-14	Allegheny County	15.3	2.5	9.2	14.6	
4	2018-01-14	Allegheny County	15.3	2.5	9.2	14.6	
...	...	...	...	...	...	...	
5303	2024-04-11	Allegheny County	68.8	58.9	61.6	68.8	
5306	2024-04-12	Allegheny County	59.4	46.8	50.8	59.4	
5307	2024-04-13	Allegheny County	59.2	44.5	51.1	59.2	
5311	2024-04-14	Allegheny County	81.1	45.0	64.1	79.9	
5312	2024-04-14	Allegheny County	81.1	45.0	64.1	79.9	

	feelslikemin	feelslike	dew	humidity	...	age_range	race	\
0	-16.6	0.6	-3.8	57.2	...	30 - 39	BLACK	
1	24.1	26.8	26.6	78.9	...	40 - 49	WHITE	
2	-6.0	3.5	0.5	68.0	...	40 - 49	WHITE	
3	-6.0	3.5	0.5	68.0	...	20 - 24	WHITE	
4	-6.0	3.5	0.5	68.0	...	30 - 39	WHITE	
...	...	...	...	...	...	...	...	
5303	58.9	61.6	56.7	84.8	...	30 - 39	BLACK	
5306	39.8	46.5	46.3	84.5	...	25 - 29	WHITE	
5307	37.4	47.9	33.0	51.5	...	20 - 24	WHITE	
5311	39.9	62.9	39.5	42.6	...	20 - 24	WHITE	
5312	39.9	62.9	39.5	42.6	...	30 - 39	BLACK	

	ethnicity_desc	accidental_exposure	victim_od_drug_id	\
0	Unknown	N	3288	
1	NON-HISPANIC OR NOT LATINO	N	527	
2	NON-HISPANIC OR NOT LATINO	N	800	
3	NON-HISPANIC OR NOT LATINO	N	804	
4	NON-HISPANIC OR NOT LATINO	N	799	
...	...	...	...	
5303	NON-HISPANIC OR NOT LATINO	N	55808	
5306	NON-HISPANIC OR NOT LATINO	N	55823	
5307	NON-HISPANIC OR NOT LATINO	N	55826	
5311	NON-HISPANIC OR NOT LATINO	N	55849	
5312	NON-HISPANIC OR NOT LATINO	N	55827	

	susp_od_drug_desc	naloxone_administered	administration_id	survive	\
0	HEROIN	N	0	Y	
1	HEROIN	Y	244	Y	
2	HEROIN	Y	454	Y	
3	HEROIN	Y	457	Y	

4	HEROIN	Y	452	Y
...	...	...	...	...
5303	FENTANYL	Y	25965	U
5306	HEROIN	Y	25969	Y
5307	HEROIN	Y	25972	Y
5311	HEROIN	Y	25981	Y
5312	HEROIN	Y	25973	U

	dose_unit
0	0
1	0
2	0
3	0
4	0
...	...
5303	2
5306	4
5307	4
5311	4
5312	4

[3684 rows x 38 columns]

3 Dropping Unneeded Columns

Now that we've removed duplicates, we can drop out unneeded columns

```
[ ]: # Columns to be dropping as they have no relevance to our work
drop_cols = ['name', 'moonphase', 'icon', 'dose_count', 'incident_id',
             ↪ 'incident_time',
             'day', 'incident_county_name', 'incident_state', 'victim_id',
             'gender_desc', 'age_range', 'race', 'ethnicity_desc',
             ↪ 'accidental_exposure',
             'victim_od_drug_id', 'susp_od_drug_desc', 'naloxone_administered',
             'administration_id', 'dose_unit']
```

```
[ ]: # Drop the drop_cols along the feature axis
df3 = df2.drop(drop_cols, axis=1)
df3
```

[]:	date	tempmax	tempmin	temp	feelslikemax	feelslikemin	\
0	2018-01-07	26.2	-4.2	10.1	18.1	-16.6	
1	2018-01-09	35.8	28.0	32.5	34.7	24.1	
2	2018-01-14	15.3	2.5	9.2	14.6	-6.0	
3	2018-01-14	15.3	2.5	9.2	14.6	-6.0	
4	2018-01-14	15.3	2.5	9.2	14.6	-6.0	
...	...	...	...	...	...	...	

5303	2024-04-11	68.8	58.9	61.6	68.8	58.9
5306	2024-04-12	59.4	46.8	50.8	59.4	39.8
5307	2024-04-13	59.2	44.5	51.1	59.2	37.4
5311	2024-04-14	81.1	45.0	64.1	79.9	39.9
5312	2024-04-14	81.1	45.0	64.1	79.9	39.9

	feelslike	dew	humidity	precip	precipcover	preciptype	snow \
0	0.6	-3.8	57.2	0.000	0.00	none	0.0
1	26.8	26.6	78.9	0.000	0.00	none	0.0
2	3.5	0.5	68.0	0.000	0.00	none	0.0
3	3.5	0.5	68.0	0.000	0.00	none	0.0
4	3.5	0.5	68.0	0.000	0.00	none	0.0
...	...	...	...	...	...	...	...
5303	61.6	56.7	84.8	1.526	83.33	rain	0.0
5306	46.5	46.3	84.5	0.266	83.33	rain	0.0
5307	47.9	33.0	51.5	0.019	12.50	rain	0.0
5311	62.9	39.5	42.6	0.001	4.17	rain	0.0
5312	62.9	39.5	42.6	0.001	4.17	rain	0.0

	snowdepth	cloudcover	uvindex	severerisk	survive
0	0.1	43.9	2	0.0	Y
1	0.1	66.3	2	0.0	Y
2	2.4	20.4	2	0.0	Y
3	2.4	20.4	2	0.0	Y
4	2.4	20.4	2	0.0	Y
...	...	...	...	...	...
5303	0.0	95.1	3	10.0	U
5306	0.0	100.0	2	10.0	Y
5307	0.0	54.5	9	10.0	Y
5311	0.0	25.2	8	30.0	Y
5312	0.0	25.2	8	30.0	U

[3684 rows x 18 columns]

4 Converting to Numbered Days

The date is non-cyclical, it simply increases. We need to get which numeric date in the year it is (1 - 365)

```
[ ]: # All the dates are in strings
type(df3['date'][0])
```

```
[ ]: str
```

```
[ ]: # Convert the string to a pandas datetime
df3['date'] = pd.to_datetime(df3['date'])
```

```
[ ]: # Double check that they are datetimes
type(df3['date'][0])
```

```
[ ]: pandas._libs.tslibs.timestamps.Timestamp
```

```
[ ]: # Convert the pandas datetime to the day of the year (1 - 365)
df3['num_date'] = df3['date'].dt.day_of_year
```

```
[ ]: # Check the output
df3
```

```
[ ]:
      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0  2018-01-07    26.2    -4.2   10.1         18.1        -16.6
1  2018-01-09    35.8    28.0   32.5         34.7         24.1
2  2018-01-14    15.3     2.5    9.2         14.6         -6.0
3  2018-01-14    15.3     2.5    9.2         14.6         -6.0
4  2018-01-14    15.3     2.5    9.2         14.6         -6.0
...
5303 2024-04-11    68.8    58.9   61.6         68.8         58.9
5306 2024-04-12    59.4    46.8   50.8         59.4         39.8
5307 2024-04-13    59.2    44.5   51.1         59.2         37.4
5311 2024-04-14    81.1    45.0   64.1         79.9         39.9
5312 2024-04-14    81.1    45.0   64.1         79.9         39.9
```

```

      feelslike  dew  humidity  precip  precipcover  preciptype  snow  \
0         0.6  -3.8     57.2   0.000         0.00      none   0.0
1        26.8  26.6     78.9   0.000         0.00      none   0.0
2         3.5   0.5     68.0   0.000         0.00      none   0.0
3         3.5   0.5     68.0   0.000         0.00      none   0.0
4         3.5   0.5     68.0   0.000         0.00      none   0.0
...
5303        61.6  56.7     84.8   1.526        83.33      rain   0.0
5306        46.5  46.3     84.5   0.266        83.33      rain   0.0
5307        47.9  33.0     51.5   0.019        12.50      rain   0.0
5311        62.9  39.5     42.6   0.001         4.17      rain   0.0
5312        62.9  39.5     42.6   0.001         4.17      rain   0.0
```

```

      snowdepth  cloudcover  uvindex  severerisk  survive  num_date
0         0.1         43.9         2         0.0         Y         7
1         0.1         66.3         2         0.0         Y         9
2         2.4         20.4         2         0.0         Y        14
3         2.4         20.4         2         0.0         Y        14
4         2.4         20.4         2         0.0         Y        14
...
5303         0.0         95.1         3        10.0         U       102
5306         0.0        100.0         2        10.0         Y       103
5307         0.0         54.5         9        10.0         Y       104
```

5311	0.0	25.2	8	30.0	Y	105
5312	0.0	25.2	8	30.0	U	105

[3684 rows x 19 columns]

5 One-hot Encoding

There is only one categorical variable now, the `preciptype`. This is also strangely listed as multiple things ‘rain, snow, ice’.

How should we encode this?

```
[ ]: df3['preciptype'].unique()

[ ]: array(['none', 'rain', 'rain,snow', 'snow', 'rain,snow,ice',
          'rain,freezingrain', 'freezingrain,snow,ice',
          'rain,freezingrain,snow', 'freezingrain', 'freezingrain,snow'],
         dtype=object)
```

One possible option is to just take every different unique value and make it a one-hot.

We can see from the `value_counts()` that there are not very many data points that have doubled up th tags with the most common being ‘rain, snow’.

```
[ ]: df3['preciptype'].value_counts()
```

```
[ ]: precipriptype
rain                1761
none                1373
rain,snow           379
snow                126
rain,freezingrain   12
freezingrain,snow   10
freezingrain         9
rain,snow,ice        6
freezingrain,snow,ice 4
rain,freezingrain,snow 4
Name: count, dtype: int64
```

```
[ ]: df4 = df3.copy()
```

```
[ ]: df4['preciptype'] = df4['preciptype'].astype('category')
df4.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3684 entries, 0 to 5312
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
#   Column                Non-Null Count  Dtype
```

```

0   date          3684 non-null  datetime64[ns]
1   tempmax       3684 non-null  float64
2   tempmin       3684 non-null  float64
3   temp          3684 non-null  float64
4   feelslikemax  3684 non-null  float64
5   feelslikemin  3684 non-null  float64
6   feelslike     3684 non-null  float64
7   dew           3684 non-null  float64
8   humidity      3684 non-null  float64
9   precip        3684 non-null  float64
10  precipcover   3684 non-null  float64
11  preciptype    3684 non-null  category
12  snow          3684 non-null  float64
13  snowdepth     3684 non-null  float64
14  cloudcover    3684 non-null  float64
15  uvindex       3684 non-null  int64
16  severerisk    3684 non-null  float64
17  survive       3684 non-null  object
18  num_date      3684 non-null  int32
dtypes: category(1), datetime64[ns](1), float64(14), int32(1), int64(1),
object(1)
memory usage: 665.5+ KB

```

```
[ ]: df4
```

```

[ ]:
      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0   2018-01-07    26.2    -4.2   10.1          18.1          -16.6
1   2018-01-09    35.8    28.0   32.5          34.7           24.1
2   2018-01-14    15.3     2.5    9.2          14.6           -6.0
3   2018-01-14    15.3     2.5    9.2          14.6           -6.0
4   2018-01-14    15.3     2.5    9.2          14.6           -6.0
...
5303 2024-04-11    68.8    58.9   61.6          68.8           58.9
5306 2024-04-12    59.4    46.8   50.8          59.4           39.8
5307 2024-04-13    59.2    44.5   51.1          59.2           37.4
5311 2024-04-14    81.1    45.0   64.1          79.9           39.9
5312 2024-04-14    81.1    45.0   64.1          79.9           39.9

      feelslike  dew  humidity  precip  precipcover  preciptype  snow  \
0           0.6 -3.8     57.2   0.000          0.00       none   0.0
1          26.8 26.6     78.9   0.000          0.00       none   0.0
2           3.5  0.5     68.0   0.000          0.00       none   0.0
3           3.5  0.5     68.0   0.000          0.00       none   0.0
4           3.5  0.5     68.0   0.000          0.00       none   0.0
...
5303          61.6 56.7     84.8   1.526          83.33       rain   0.0
5306          46.5 46.3     84.5   0.266          83.33       rain   0.0

```

5307	47.9	33.0	51.5	0.019	12.50	rain	0.0
5311	62.9	39.5	42.6	0.001	4.17	rain	0.0
5312	62.9	39.5	42.6	0.001	4.17	rain	0.0

	snowdepth	cloudcover	uvindex	severerisk	survive	num_date
0	0.1	43.9	2	0.0	Y	7
1	0.1	66.3	2	0.0	Y	9
2	2.4	20.4	2	0.0	Y	14
3	2.4	20.4	2	0.0	Y	14
4	2.4	20.4	2	0.0	Y	14
...	...	...	...	...	...	...
5303	0.0	95.1	3	10.0	U	102
5306	0.0	100.0	2	10.0	Y	103
5307	0.0	54.5	9	10.0	Y	104
5311	0.0	25.2	8	30.0	Y	105
5312	0.0	25.2	8	30.0	U	105

[3684 rows x 19 columns]

```
[ ]: df4.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3684 entries, 0 to 5312
Data columns (total 19 columns):
#   Column          Non-Null Count  Dtype
---  -
0   date             3684 non-null  datetime64[ns]
1   tempmax          3684 non-null  float64
2   tempmin          3684 non-null  float64
3   temp             3684 non-null  float64
4   feelslikemax     3684 non-null  float64
5   feelslikemin     3684 non-null  float64
6   feelslike        3684 non-null  float64
7   dew              3684 non-null  float64
8   humidity         3684 non-null  float64
9   precip           3684 non-null  float64
10  precipcover      3684 non-null  float64
11  preciptype       3684 non-null  category
12  snow             3684 non-null  float64
13  snowdepth        3684 non-null  float64
14  cloudcover       3684 non-null  float64
15  uvindex          3684 non-null  int64
16  severerisk       3684 non-null  float64
17  survive          3684 non-null  object
18  num_date         3684 non-null  int32
dtypes: category(1), datetime64[ns](1), float64(14), int32(1), int64(1),
object(1)
memory usage: 665.5+ KB
```

```
[ ]: df4['survive'] = df4['survive'].replace('U', np.NaN)
df4['survive'] = df4['survive'].map({'Y':1, 'N':0})
```

```
[ ]: df4
```

```
[ ]:
      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0  2018-01-07    26.2    -4.2  10.1         18.1        -16.6
1  2018-01-09    35.8    28.0  32.5         34.7         24.1
2  2018-01-14    15.3     2.5   9.2         14.6         -6.0
3  2018-01-14    15.3     2.5   9.2         14.6         -6.0
4  2018-01-14    15.3     2.5   9.2         14.6         -6.0
...
5303 2024-04-11    68.8    58.9  61.6         68.8         58.9
5306 2024-04-12    59.4    46.8  50.8         59.4         39.8
5307 2024-04-13    59.2    44.5  51.1         59.2         37.4
5311 2024-04-14    81.1    45.0  64.1         79.9         39.9
5312 2024-04-14    81.1    45.0  64.1         79.9         39.9

      feelslike  dew  humidity  precip  precipcover  preciptype  snow  \
0           0.6 -3.8      57.2   0.000         0.00      none   0.0
1          26.8 26.6      78.9   0.000         0.00      none   0.0
2           3.5  0.5      68.0   0.000         0.00      none   0.0
3           3.5  0.5      68.0   0.000         0.00      none   0.0
4           3.5  0.5      68.0   0.000         0.00      none   0.0
...
5303        61.6 56.7      84.8   1.526        83.33      rain   0.0
5306        46.5 46.3      84.5   0.266        83.33      rain   0.0
5307        47.9 33.0      51.5   0.019        12.50      rain   0.0
5311        62.9 39.5      42.6   0.001         4.17      rain   0.0
5312        62.9 39.5      42.6   0.001         4.17      rain   0.0

      snowdepth  cloudcover  uvindex  severerisk  survive  num_date
0           0.1         43.9        2         0.0        1.0         7
1           0.1         66.3        2         0.0        1.0         9
2           2.4         20.4        2         0.0        1.0        14
3           2.4         20.4        2         0.0        1.0        14
4           2.4         20.4        2         0.0        1.0        14
...
5303          0.0         95.1        3        10.0        NaN        102
5306          0.0        100.0        2        10.0        1.0        103
5307          0.0         54.5        9        10.0        1.0        104
5311          0.0         25.2        8        30.0        1.0        105
5312          0.0         25.2        8        30.0        NaN        105
```

```
[3684 rows x 19 columns]
```

```
[ ]: df4.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 3684 entries, 0 to 5312
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   date                   3684 non-null   datetime64[ns]
1   tempmax                3684 non-null   float64
2   tempmin                3684 non-null   float64
3   temp                  3684 non-null   float64
4   feelslikemax           3684 non-null   float64
5   feelslikemin           3684 non-null   float64
6   feelslike              3684 non-null   float64
7   dew                    3684 non-null   float64
8   humidity               3684 non-null   float64
9   precip                 3684 non-null   float64
10  precipcover            3684 non-null   float64
11  preciptype             3684 non-null   category
12  snow                   3684 non-null   float64
13  snowdepth              3684 non-null   float64
14  cloudcover             3684 non-null   float64
15  uvindex                3684 non-null   int64
16  severerisk             3684 non-null   float64
17  survive                3486 non-null   float64
18  num_date               3684 non-null   int32
dtypes: category(1), datetime64[ns](1), float64(15), int32(1), int64(1)
memory usage: 665.5 KB

```

```

[ ]: df5 = pd.get_dummies(df4, dtype=int)
df5

```

```

[ ]:
      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0  2018-01-07    26.2    -4.2  10.1          18.1         -16.6
1  2018-01-09    35.8    28.0  32.5          34.7          24.1
2  2018-01-14    15.3     2.5   9.2          14.6          -6.0
3  2018-01-14    15.3     2.5   9.2          14.6          -6.0
4  2018-01-14    15.3     2.5   9.2          14.6          -6.0
...
5303  2024-04-11    68.8    58.9  61.6          68.8          58.9
5306  2024-04-12    59.4    46.8  50.8          59.4          39.8
5307  2024-04-13    59.2    44.5  51.1          59.2          37.4
5311  2024-04-14    81.1    45.0  64.1          79.9          39.9
5312  2024-04-14    81.1    45.0  64.1          79.9          39.9

      feelslike  dew  humidity  precip  ...  preciptype_freezingrain  \
0           0.6 -3.8     57.2   0.000  ...                        0
1          26.8 26.6     78.9   0.000  ...                        0
2           3.5  0.5     68.0   0.000  ...                        0

```

3	3.5	0.5	68.0	0.000	...	0
4	3.5	0.5	68.0	0.000	...	0
...	...	...	...	...	...	...
5303	61.6	56.7	84.8	1.526	...	0
5306	46.5	46.3	84.5	0.266	...	0
5307	47.9	33.0	51.5	0.019	...	0
5311	62.9	39.5	42.6	0.001	...	0
5312	62.9	39.5	42.6	0.001	...	0

	preciptype_freezingrain,snow	preciptype_freezingrain,snow,ice	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
5303	0	0	
5306	0	0	
5307	0	0	
5311	0	0	
5312	0	0	

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1	0	0	
1	1	0	0	
2	1	0	0	
3	1	0	0	
4	1	0	0	
...	...	...	...	
5303	0	1	0	
5306	0	1	0	
5307	0	1	0	
5311	0	1	0	
5312	0	1	0	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
5303	0	0	
5306	0	0	
5307	0	0	
5311	0	0	
5312	0	0	

	preciptype_rain,snow,ice	preciptype_snow
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0
...	...	...
5303	0	0
5306	0	0
5307	0	0
5311	0	0
5312	0	0

[3684 rows x 28 columns]

The other method would be to search each string for values and then for doubled-up values, 'rain, snow', we would set both rain and snow as true, rather than having 'rain, snow' set to true.

The downside to this is that 'rain, snow' correlates well to a temperature region of around 30 - 35 F, which correlates to a specific time of year. You may not get this kind of inference from marking rain and snow individually.

```
[ ]: df5 = df5.dropna()
df5
```

```
[ ]:
      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0  2018-01-07    26.2    -4.2  10.1         18.1        -16.6
1  2018-01-09    35.8    28.0  32.5         34.7         24.1
2  2018-01-14    15.3     2.5   9.2         14.6         -6.0
3  2018-01-14    15.3     2.5   9.2         14.6         -6.0
4  2018-01-14    15.3     2.5   9.2         14.6         -6.0
...  ...  ...  ...  ...  ...  ...
5299 2024-04-08    73.0    49.3  60.1         73.0         47.3
5301 2024-04-10    65.1    57.1  60.6         65.1         57.1
5306 2024-04-12    59.4    46.8  50.8         59.4         39.8
5307 2024-04-13    59.2    44.5  51.1         59.2         37.4
5311 2024-04-14    81.1    45.0  64.1         79.9         39.9
```

	feelslike	dew	humidity	precip	...	preciptype_freezingrain	\
0	0.6	-3.8	57.2	0.000	...		0
1	26.8	26.6	78.9	0.000	...		0
2	3.5	0.5	68.0	0.000	...		0
3	3.5	0.5	68.0	0.000	...		0
4	3.5	0.5	68.0	0.000	...		0
...	...	...	...	...	...	...	
5299	60.0	37.1	42.8	0.006	...		0
5301	60.6	54.5	80.9	0.179	...		0

5306	46.5	46.3	84.5	0.266	...	0
5307	47.9	33.0	51.5	0.019	...	0
5311	62.9	39.5	42.6	0.001	...	0

	preciptype_freezingrain,snow	preciptype_freezingrain,snow,ice	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
5299	0	0	
5301	0	0	
5306	0	0	
5307	0	0	
5311	0	0	

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1	0	0	
1	1	0	0	
2	1	0	0	
3	1	0	0	
4	1	0	0	
...	...	...	...	
5299	0	1	0	
5301	0	1	0	
5306	0	1	0	
5307	0	1	0	
5311	0	1	0	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
5299	0	0	
5301	0	0	
5306	0	0	
5307	0	0	
5311	0	0	

	preciptype_rain,snow,ice	preciptype_snow
0	0	0
1	0	0
2	0	0

3	0	0
4	0	0
...	...	...
5299	0	0
5301	0	0
5306	0	0
5307	0	0
5311	0	0

[3486 rows x 28 columns]

6 Reindex

```
[ ]: df6 = df5.reset_index(drop=True).copy()
df6
```

```
[ ]:
      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0  2018-01-07    26.2    -4.2  10.1         18.1        -16.6
1  2018-01-09    35.8    28.0  32.5         34.7         24.1
2  2018-01-14    15.3     2.5   9.2         14.6         -6.0
3  2018-01-14    15.3     2.5   9.2         14.6         -6.0
4  2018-01-14    15.3     2.5   9.2         14.6         -6.0
...  ...  ...  ...  ...  ...  ...
3481 2024-04-08    73.0    49.3  60.1         73.0         47.3
3482 2024-04-10    65.1    57.1  60.6         65.1         57.1
3483 2024-04-12    59.4    46.8  50.8         59.4         39.8
3484 2024-04-13    59.2    44.5  51.1         59.2         37.4
3485 2024-04-14    81.1    45.0  64.1         79.9         39.9

      feelslike  dew  humidity  precip  ...  precip_type_freezingrain  \
0           0.6 -3.8     57.2   0.000  ...                0
1          26.8 26.6     78.9   0.000  ...                0
2           3.5  0.5     68.0   0.000  ...                0
3           3.5  0.5     68.0   0.000  ...                0
4           3.5  0.5     68.0   0.000  ...                0
...  ...  ...  ...  ...  ...
3481        60.0 37.1     42.8   0.006  ...                0
3482        60.6 54.5     80.9   0.179  ...                0
3483        46.5 46.3     84.5   0.266  ...                0
3484        47.9 33.0     51.5   0.019  ...                0
3485        62.9 39.5     42.6   0.001  ...                0

      precip_type_freezingrain,snow  precip_type_freezingrain,snow,ice  \
0                                0                                0
1                                0                                0
2                                0                                0
```

3	0	0
4	0	0
...	...	...
3481	0	0
3482	0	0
3483	0	0
3484	0	0
3485	0	0

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1	0	0	
1	1	0	0	
2	1	0	0	
3	1	0	0	
4	1	0	0	
...	...	...	...	
3481	0	1	0	
3482	0	1	0	
3483	0	1	0	
3484	0	1	0	
3485	0	1	0	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
3481	0	0	
3482	0	0	
3483	0	0	
3484	0	0	
3485	0	0	

	preciptype_rain,snow,ice	preciptype_snow
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0
...	...	...
3481	0	0
3482	0	0
3483	0	0
3484	0	0
3485	0	0

[3486 rows x 28 columns]

7 Drop the date

I don't know if this needs to be dropped, we may need the years later if trying to plot the data...

```
[ ]: df7 = df6.drop('date', axis=1)
df7
```

```
[ ]:      tempmax  tempmin  temp  feelslikemax  feelslikemin  feelslike  dew  \
0         26.2     -4.2  10.1         18.1         -16.6         0.6  -3.8
1         35.8     28.0  32.5         34.7          24.1        26.8  26.6
2         15.3       2.5   9.2         14.6          -6.0         3.5   0.5
3         15.3       2.5   9.2         14.6          -6.0         3.5   0.5
4         15.3       2.5   9.2         14.6          -6.0         3.5   0.5
...      ...      ...  ...      ...      ...      ...  ...
3679      68.8     58.9  61.6         68.8         58.9        61.6  56.7
3680      59.4     46.8  50.8         59.4         39.8        46.5  46.3
3681      59.2     44.5  51.1         59.2         37.4        47.9  33.0
3682      81.1     45.0  64.1         79.9         39.9        62.9  39.5
3683      81.1     45.0  64.1         79.9         39.9        62.9  39.5
```

```
      humidity  precip  precipcover  ...  precip_type_freezingrain  \
0         57.2    0.000         0.00  ...                          0
1         78.9    0.000         0.00  ...                          0
2         68.0    0.000         0.00  ...                          0
3         68.0    0.000         0.00  ...                          0
4         68.0    0.000         0.00  ...                          0
...      ...      ...      ...      ...
3679      84.8    1.526         83.33  ...                          0
3680      84.5    0.266         83.33  ...                          0
3681      51.5    0.019         12.50  ...                          0
3682      42.6    0.001          4.17  ...                          0
3683      42.6    0.001          4.17  ...                          0
```

```
      precip_type_freezingrain,snow  precip_type_freezingrain,snow,ice  \
0                                0                                0
1                                0                                0
2                                0                                0
3                                0                                0
4                                0                                0
...      ...      ...
3679      0                                0
3680      0                                0
3681      0                                0
3682      0                                0
```

3683		0	0
	preciptype_none	preciptype_rain	preciptype_rain,freezingrain \
0	1	0	0
1	1	0	0
2	1	0	0
3	1	0	0
4	1	0	0
...	...	...	...
3679	0	1	0
3680	0	1	0
3681	0	1	0
3682	0	1	0
3683	0	1	0

	preciptype_rain,freezingrain,snow	preciptype_rain,snow \
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0
...	...	...
3679	0	0
3680	0	0
3681	0	0
3682	0	0
3683	0	0

	preciptype_rain,snow,ice	preciptype_snow
0	0	0
1	0	0
2	0	0
3	0	0
4	0	0
...	...	...
3679	0	0
3680	0	0
3681	0	0
3682	0	0
3683	0	0

[3684 rows x 26 columns]

8 Write-out dataset

```
[ ]: os.getcwd()

[ ]: 'C:\\Users\\btb51\\Documents\\GitHub\\DAAN881_Team1\\opioid_analytics\\data\\processed'

[ ]: os.chdir(path='C:
↪\\Users\\btb51\\Documents\\GitHub\\DAAN881_Team1\\opioid_analytics\\data\\processed\\')

[ ]: df5.to_csv('cleaned_data_survive_withIndex.csv', index=True)

[ ]: df5[['tempmax', 'survive']].head(15)

[ ]:
    tempmax  survive
0      26.2      1.0
1      35.8      1.0
2      15.3      1.0
3      15.3      1.0
4      15.3      1.0
5      55.1      1.0
6      47.2      1.0
9      42.9      1.0
10     44.2      0.0
11     55.8      0.0
16     62.0      0.0
17     70.6      1.0
18     41.5      1.0
19     57.8      1.0
20     51.3      1.0

[ ]: df6.to_csv('cleaned_data_survive.csv', index=False)

[ ]: df7.to_csv('../data/processed/cleaned_data.csv', index=False)

[ ]:
```